

STD strength matching between one and two depression timescales (2026-09-13)

Decision record for the paper's single- vs multiple-timescale comparison. Equations and the authoritative statement: docs/EquationsParametersDocs/Equations_stability_paper.md, § "2026-09-13: STD strength matching". Presets: srnn_param_presetcases celltype_pairs_sfaEI_Sc0p2sig0p1_noise0p025_dualStdScaled_3cond_mu8p25 and ..._dualStdUsage_3cond_mu8p25. Test: scripts/tests/test_route_scale.m. Steady-state curves over the occupied rates: fig_STD_steady_state.

1. The decision

Made by TR, 2026-09-13, on the plan for the overnight of 2026-09-13/14.

The paper compares no adaptation, one-timescale adaptation (1 SFA + 1 STD) and multiple-timescale adaptation (3 SFA + 2 STD) and attributes the differences to the *temporal structure* of adaptation. SFA is normalised for that comparison: $c_{\text{eff}} = c/K$ makes the steady-state adaptation $c \cdot r$ whatever K is. STD was not. With both depression timescales at $\rho = \tau_{\text{rel}}/\tau_{\text{rec}} = 0.125$ ($\tau_{\text{rec}} [2\ 4]$ s, $\tau_{\text{rel}} [0.25\ 0.5]$ s) and the single-timescale routes at (2, 0.25), the two-timescale steady state $1/(1 + r/\rho)^2$ is the **square** of the one-timescale $1/(1 + r/\rho)$, so adding a timescale also multiplied the depression. Codex's audit (scripts/paper/fig_to_do_in_future.md § 1) and StochasticPlasticDynamicalSystemPaper/reports/suggest_updates_main_matlab_model.md both flagged this as the model-defining decision to freeze before any final run.

What is matched: the steady-state synaptic output $\theta_{\text{ss}}(r) = r \cdot \Pi_m b_m(r)$ of the two-timescale routes equals that of the one-timescale routes at a reference rate r_{ref} , chosen by rule as the occupied median rate (below).

How (primary): a multiplicative scale on the two-timescale routes. The depression variables, θ and the ODE are untouched; the weights of the four sfa3_std2 routes are multiplied by

$$s = \theta_{\text{single}}(r_{\text{ref}}) / \theta_{\text{dual}}(r_{\text{ref}}) = 1 + r_{\text{ref}}/\rho = 3.$$

Implemented as `synapse_config.<pre>.<post>.scale`, applied to `params.W` in `SRNNCellTypePairs.get_params` only (the drawn W stays unscaled; dynamics, both Jacobians and `jacobian_times` read `params.W`, so they agree by construction; the ESN 'synaptic' readout's route-redundancy check compares the scale too). `no_adaptation` and `sfa1_std1` are bit-identical to the unmatched preset's.

Direction: scale the dual condition *up* to the single one ($s = 3$ on dual, 1 on single), so the single-timescale condition stays exactly the current network. Alternatives offered and not taken: single scaled down to dual ($s \approx 0.33$), or both adapted conditions scaled to no adaptation ($s = 3$ and 9).

Control (also run): the usage-matched preset, which needs no code change.

2. The alternative that was raised and overruled

The recommendation put to TR was to match by **usage**: keep $\tau_{\text{rec}} = [2\ 4]$ s and choose τ_{rel} so the product equals the single factor at r_{ref} . With equal usage ρ_u on both timescales,

$$(1 + r_{\text{ref}}/\rho_u)^2 = 1 + r_{\text{ref}}/\rho \rightarrow \rho_u = r_{\text{ref}} / (\sqrt{1 + r_{\text{ref}}/\rho} - 1) = 0.25/(\sqrt{3} - 1) = 0.34151$$
$$\tau_{\text{rel}} = \rho_u \cdot \tau_{\text{rec}} = [0.68301, 1.36603] \text{ s}.$$

Reasons it was recommended: τ_{rel} is Varela's release strength d_m , so this is the tuning the product model itself offers; it changes no code; and it leaves the **undepressed (low-rate) recurrent gain identical across the three conditions**. Its cost: each timescale depresses less and the depression dynamics are slower (rate $1/\tau_{\text{rec}} + r/\tau_{\text{rel}}$ per variable).

The concern with the scale, stated to TR before the choice: because $b_m \rightarrow 1$ as $r \rightarrow 0$, the scaled dual condition runs at **3× the undepressed recurrent gain** of the other two conditions. The match holds at the operating point, not at low rates, and a change in effective W scale is itself a change the paper's `level_of_chaos` sweep treats as a stability axis. TR reaffirmed the scale as primary and asked for the usage variant as the control, so the morning comparison (§ 5) shows the consequence rather than arguing it.

3. The numbers

r_ref = 0.25. Rule: the median mean firing rate of the multiple-timescale condition at the default point of the 1-D sweeps of the medium run data/topk_med (top-K, 15 reps per level), rounded to 0.05. Per sweep, the level nearest the preset default:

sweep	sfa3_std2 median rate
f_E	0.236
level_of_chaos	0.223
mu_EE_relative	0.239
mu_IE_relative	0.251
n	0.231

(mu_EI_relative and mu_II_relative were excluded: their default is negative and the nearest-level lookup landed on the wrong sign.) Pooled over the 105 near-default reps:

condition	median	p5	p25	p75	p95
no_adaptation	0.483	0.027	0.274	0.729	0.915
sfa1_std1	0.269	0.033	0.167	0.432	0.791
sfa3_std2	0.237	0.050	0.220	0.249	0.346

The joint 64-point parameter-space sample, which ranges far from the default, has medians 0.250 / 0.159 / 0.154 for the same three conditions; it was not used because it is not the operating point.

At r_ref = 0.25, p = 0.125: one factor $1/(1+2) = 1/3$; unmatched product $1/9$; scale $s = 3$; usage $\rho_u = 0.34151$, $\tau_{rel} = [0.68301, 1.36603]$ s. Both matched variants give $\theta_{ss}(0.25) = 0.25/3 = 0.0833$, against 0.0278 unmatched.

quantity at r_ref	single (1 TS)	dual unmatched	dual scaled	dual usage
Πb_m	0.333	0.111	0.111	0.333
$\theta_{ss} = s \cdot r \cdot \Pi b_m$	0.0833	0.0278	0.0833	0.0833
undepressed gain factor ($r \rightarrow 0$)	1	1	3	1

4. Replacement manuscript text

The manuscript repo is not edited here; this is ready to paste into Manuscript5.md.

4a. What it says now (Supplemental Methods, “Short-term depression”, ~line 375)

Unlike adaptation, depression is deliberately not normalized by its number of timescales: it enters as a product rather than a sum, each factor rests at one, and adding a timescale is meant to deepen depression rather than redistribute it, following the product-of-components description of cortical depression [varelaQuantitativeDescriptionShortTerm1997].

and, earlier in the same paragraph, “so the product in {eq:theta} settles to $(1 + 8r)^{-2}$. At $r = 1$ a single timescale reduces synaptic output ninefold whereas two reduce it by a factor of 81, the square ({supFig:std_steady_state}).”

4b. Replacement paragraph (scaled variant primary, usage control)

Short-term depression. The same depression time constants were applied to all four connection routes. In the single-timescale condition the recovery time constant was eight times the release time constant, so at a constant rate r the depression variable settles to $b = 1/(1 + 8r)$. Depression enters {eq:theta} as a product of resource variables, each resting at one, following the product-of-components description of cortical depression [varelaQuantitativeDescriptionShortTerm1997]; with the same ratio on both

timescales the two-timescale product would settle to $(1 + 8r)^{-2}$, the square of the single-timescale value. Adding a depression timescale would then change the strength of depression as well as its temporal structure, and the comparison between the adapting conditions would not isolate the number of timescales. We therefore matched the steady-state synaptic output of the two-timescale routes to that of the single-timescale routes at a reference rate $r_{ref} = 0.25$, the median mean firing rate of the multiple-timescale network at the reference parameters. In the primary configuration the two-timescale routes retained their release and recovery time constants and their weights were multiplied by a route scale $s = \theta^{(1)}(r_{ref})/\theta^{(2)}(r_{ref}) = 1 + 8r_{ref} = 3$, so that the recurrent input to neuron i is $\sum_j s_{q_j \rightarrow q_i} w_{ij} \theta_j$ with $s = 3$ on all four routes of the multiple-timescale condition and $s = 1$ otherwise. This equalises synaptic output at the operating point; because the resource variables approach one at low rates, the un-depressed recurrent gain of the multiple-timescale condition is three times that of the other conditions. As a control we also ran a usage-matched configuration in which the weights were unchanged and the release time constants of the two depression variables were lengthened to $\tau_{rel} = 0.683$ and 1.366 s, giving both variables the usage ratio $\tau_{rel}/\tau_{rec} = 0.342$ for which $(1 + r_{ref}\tau_{rec}/\tau_{rel})^2 = 1 + 8r_{ref}$. The two configurations agree at r_{ref} and differ away from it ({supFig:std_steady_state}). The single-timescale and no-adaptation conditions are identical in both. Where the text attributes a difference between the adapting conditions to the number of adaptation timescales, it refers to this strength-matched comparison; the earlier unmatched configuration compared timescale count and depression strength together.

4c. Replacement rows for

Condition	τ_a (s)	τ_{rel} (s)	τ_{rec} (s)	route scale s
No Adaptation	none	none	none	1
Single-Timescale Adaptation	0.25	0.25	2	1
Multiple-Timescale Adaptation (primary)	0.25, 1.581, 10	0.25, 0.5	2, 4	3
Multiple-Timescale, usage-matched (control)	0.25, 1.581, 10	0.683, 1.366	2, 4	1

Caption addition: “The multiple-timescale routes are strength-matched to the single-timescale routes at $r_{ref} = 0.25$: in the primary configuration by a route weight scale $s = 3$, in the control by the release time constants.”

The parameters table rows for τ_{rel} (E to E ... I to I) stay [0.25 0.5] for the primary configuration; add a row “route scale s (all four routes, multiple-timescale condition) | STD strength match | 3 | –”.

5. Results: TO FILL after the fast runs

Medians over the successful jobs of each run; the unmatched column is the *medium* run (data/topk_med, 15 reps per level, T = 20 s) and the two matched columns are *fast* runs (data/stdScaled_fast, data/stdUsage_fast; fewer reps, shorter T), so compare direction and size, not decimals.

condition	measure	unmatched (topk_med, medium)	scaled (fast)	usage (fast)
no_adaptation	λ_1	TO FILL	(identical physics)	(identical physics)
no_adaptation	mean rate	TO FILL		
sfa1_std1	λ_1	TO FILL	(identical physics)	(identical physics)
sfa1_std1	mean rate	TO FILL		
sfa3_std2	λ_1	TO FILL	TO FILL	TO FILL
sfa3_std2	mean rate	TO FILL	TO FILL	TO FILL
sfa3_std2	h_KS	TO FILL	TO FILL	TO FILL
sfa3_std2	D_KY	TO FILL	TO FILL	TO FILL

Also to record: the new occupied median rate of `sfa3_std2` in each matched run (if it moves far from 0.25 the match point should be revisited once), and the `fig_STD_steady_state` matching table (θ_{ss} at `r_ref` and at the 5th/95th occupied percentiles, the maximum relative mismatch over the occupied range for both matchings).

6. Open questions for TR

1. **Direction.** Confirm scaling the dual condition up ($s = 3$) after seeing § 5. The alternative that keeps every condition at the same undepressed gain is the usage control; if its λ_1 ordering across the three conditions is the same as the scaled run's, the choice is presentational and the scale's $3\times$ low-rate gain is a caveat in the Methods; if the orderings differ, the paper's claim depends on the choice and that needs a decision.
2. **Which is primary in the manuscript.** § 4b is written with the scale as primary and usage as control, per the decision. Swapping them is a paragraph edit, not a rerun, since both are run.
3. **Does the product model stay primary?** Codex's option 2 (per-timescale exponents q_m with $\Sigma q_m = 1$) was not taken; the Methods keep the Varela product and add the match. Confirm.
4. **`r_ref` after the rerun.** `r_ref` was set from the unmatched network. If the matched network's occupied median is not ≈ 0.25 , decide whether to iterate once (the rule stays the same, the literal changes) or to keep 0.25 and report the difference.